GLMs: theoretical roadmap

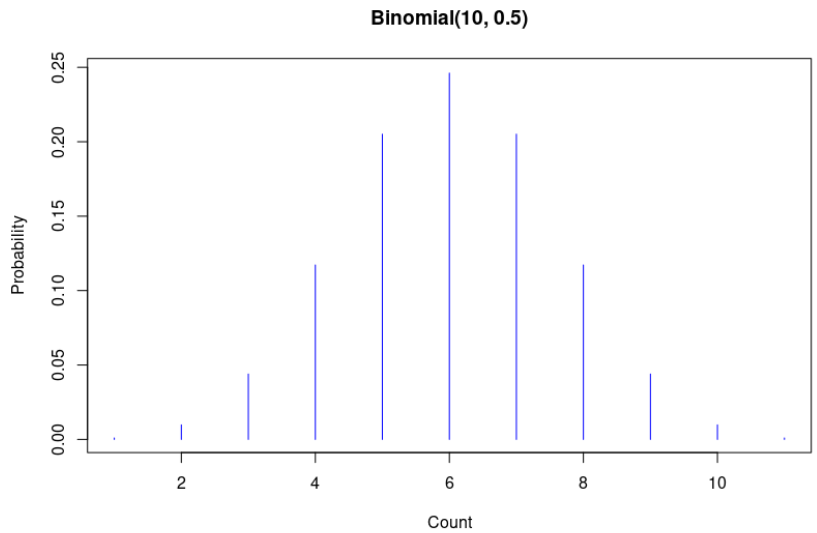
GLMs: Theoretical roadmap

- Goal: connect the linear combination of predictors to the outcome variable('s parameter(s))
- (still geocentric)
- Building strategy
  1. Pick an outcome distribution
  2. Model its (transformed) parameters using link function(s) to map them to linear models (think about priors)
  3. Compute posterior (for parameters and predictive)
- Can model multivariate relationships and non-linear responses
- Building blocks of multi-level models

Binomial distribution

- Counts y of a specific event (success) out of n possibilities
- Constant expected value
- Maximum entropy distribution is the binomial with parameters n (number of trials) and p (probability of success):

$$y \sim \text{Binomial}(n, p)$$

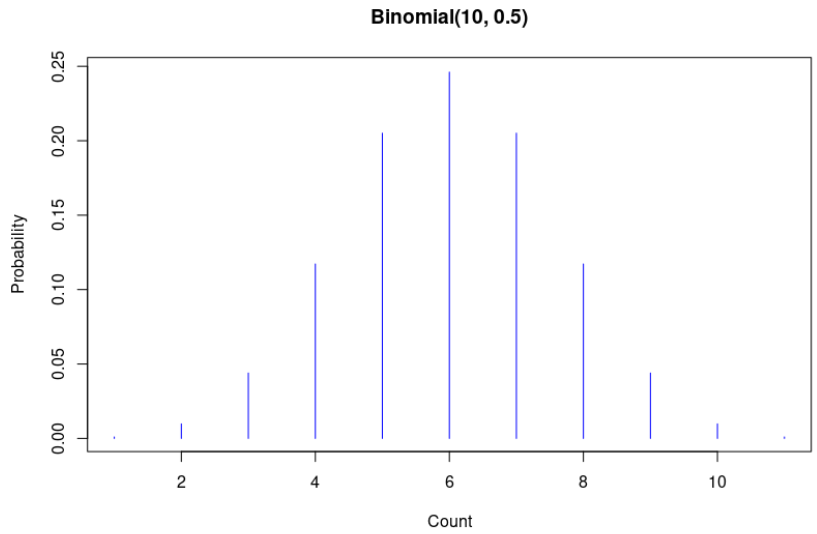


Binomial distribution

- Counts y of a specific event (success) out of n possibilities
- Constant expected value
- Maximum entropy distribution is the binomial with parameters n (number of trials) and p (probability of success):

$$\begin{aligned} y &\sim \text{Binomial}(n, p) \\ \text{E}[y] &= np \\ \text{Var}(y) &= np(1-p) \end{aligned}$$

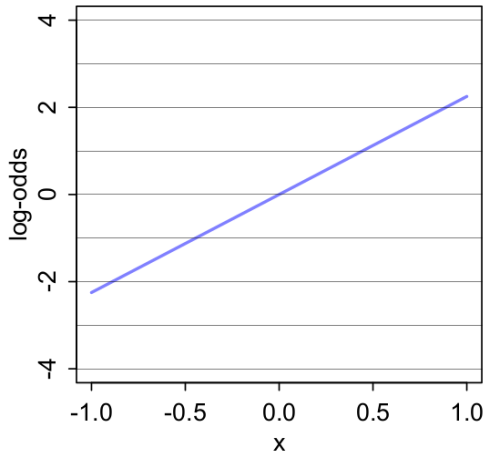
Mean and variance not independent



Binomial distribution: link function

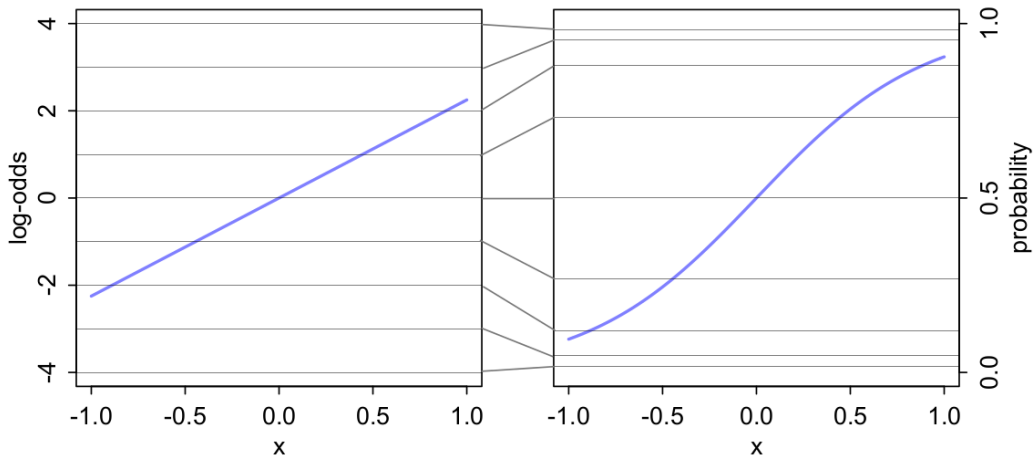
$$y \sim \text{Binomial}(n, p)$$

- Goal: map the linear model onto $[0, 1]$



Binomial distribution: logit-link function

- Goal: map the linear model onto $[0, 1]$

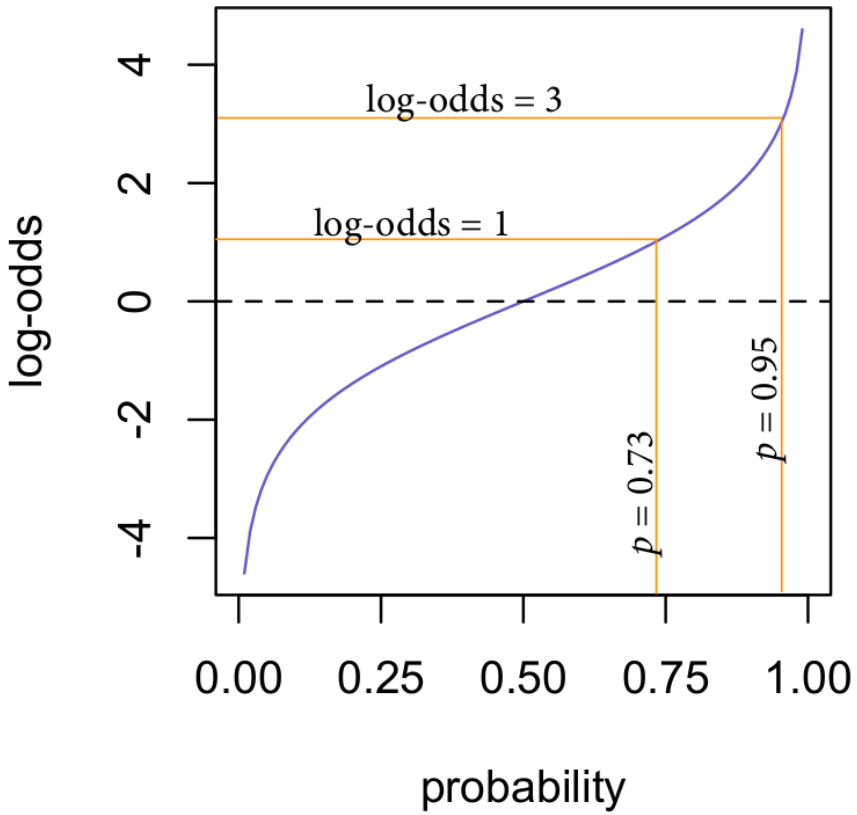


$$
\begin{aligned}
y_i &\sim \text{Binomial}(n, p_i) \\
\text{logit}(p_i) &= \alpha + \beta x
\end{aligned}
$$

Log odds

- logit- link
  $logit(p_i) = log(p_i/1 - p_i)$
- log-odds $= p_i/1 - p_i$
- inverse link $= \text{logit}^{-1}$
  $p_i = \text{logit}^{-1}(\alpha + \beta X)$
- logistic
  $log(p_i/1 - p_i = q_i$
  $p_i = exp(q_i)/1 + exp(q_i)$

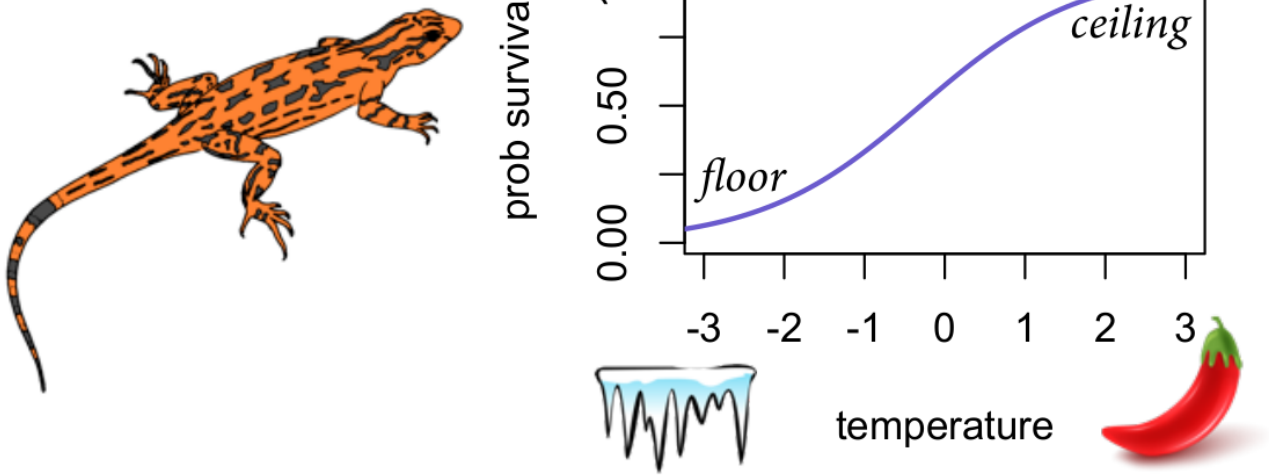
Inverse of the logit-link function: the logistic function



Other link functions: Probit, Complementary log-log.

Consequence of link functions: everything interacts

- There are *floor* and *ceiling* effects



Consequence of link functions: everything interacts

- Linear regression

$$\mu = \alpha + \beta x, \qquad \frac{\partial \mu}{\partial x} = \beta$$

while

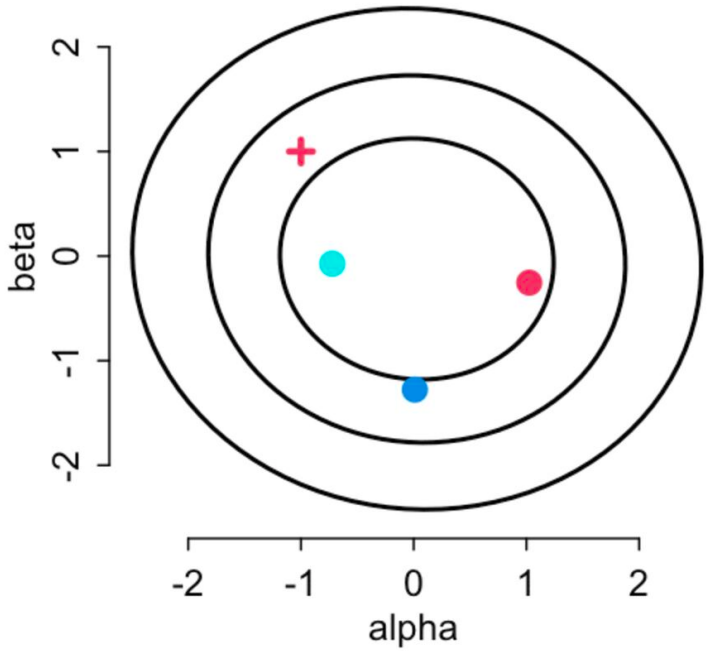
- Logistic regression

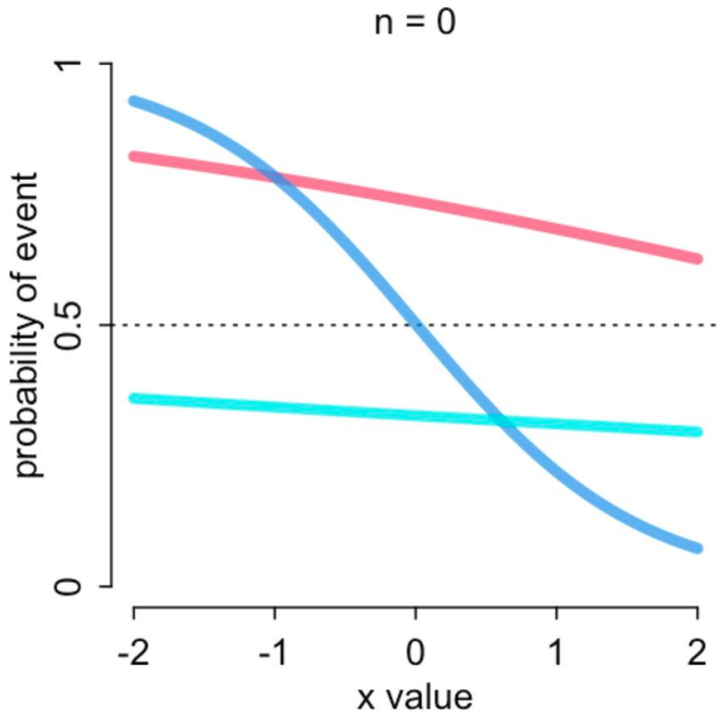
$$p = \frac{\exp(\alpha + \beta x)}{1 + \exp(\alpha + \beta x)}, \qquad \frac{\partial p}{\partial x} = \frac{\beta}{2(1 + \cosh(\alpha + \beta x))}$$

Logistic shape

$$p = \frac{\exp(\alpha+\beta x)}{1+\exp(\alpha+\beta x)}$$

$$\text{logit}(p_i) = \alpha + \beta x_i$$

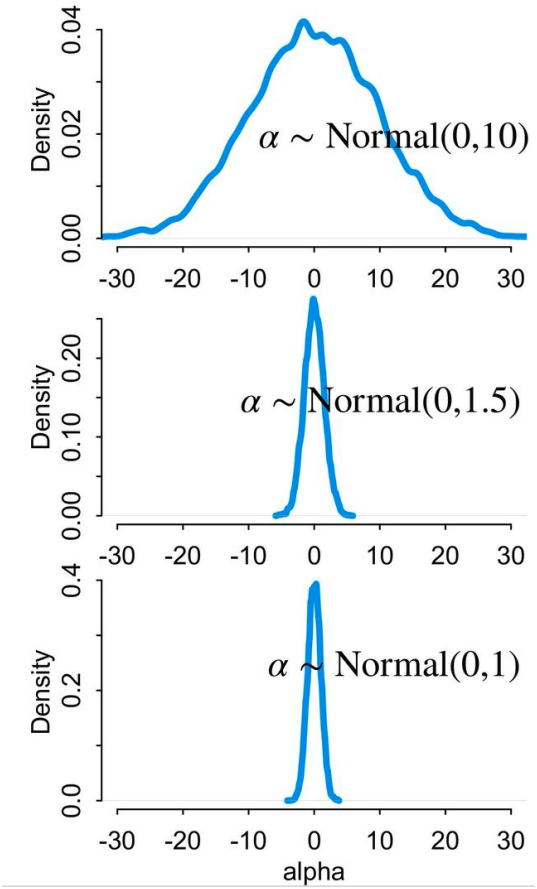




Logistic priors

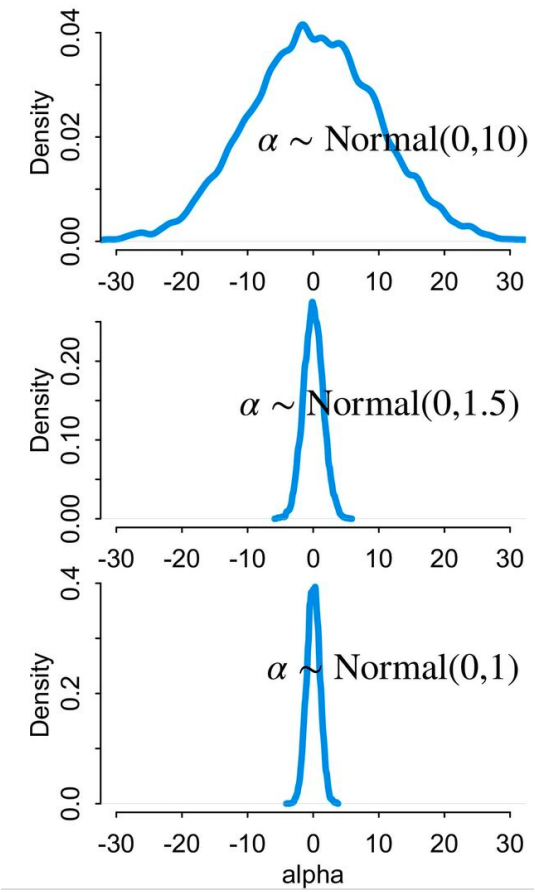
$$\text{logit}(p_i) = \alpha$$

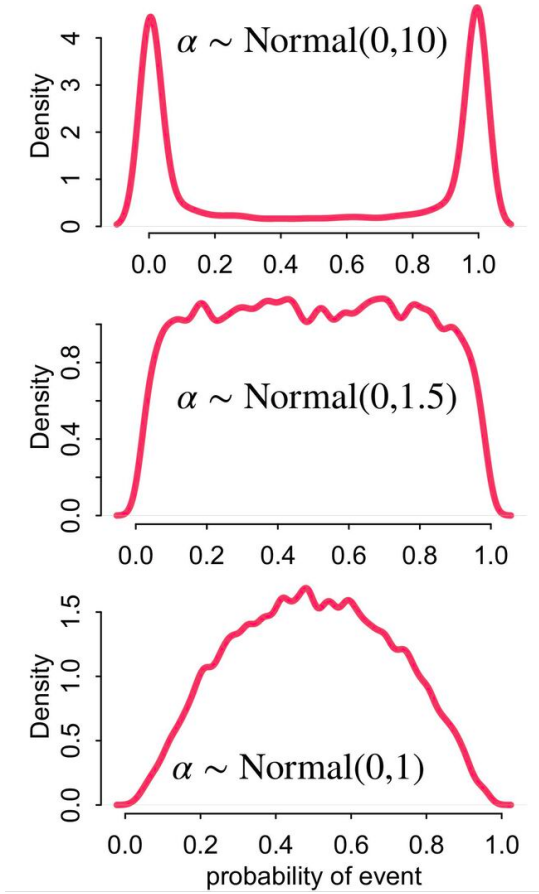
- Logit compresses the parameter distributions
- Anything above $+4$ = almost always
- Anything below -4 = almost never



Logistic priors: Probability scale

$$\text{logit}(p_i) = \alpha$$





Logistic priors

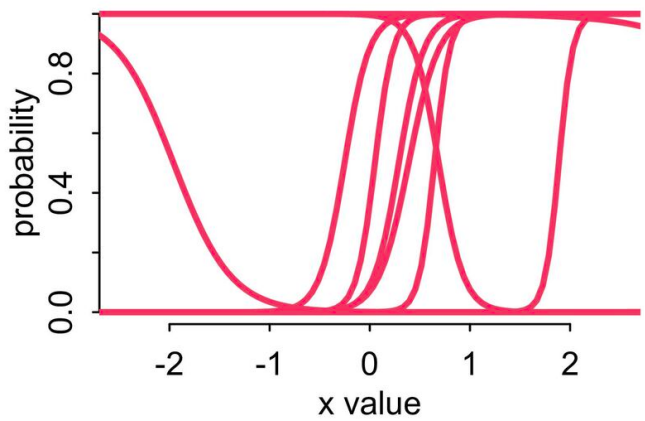
$$\text{logit}(p_i) = \alpha + \beta x_i$$

$$\alpha \sim \text{Normal}(0,10)$$
$$\beta \sim \text{Normal}(0,10)$$

```
a <- rnorm(1e4,0,10)
b <- rnorm(1e4,0,10)

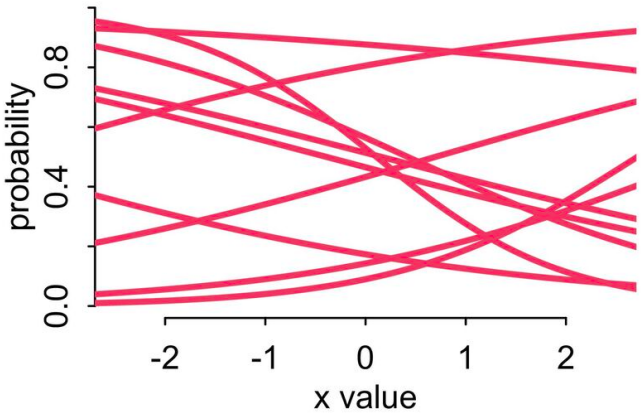
xseq <- seq(from=-3,to=3,len=100)
p <- sapply( xseq , function(x)
inv_logit(a+b*x) )

plot( NULL , xlim=c(-2.5,2.5) , ylim=c(0,1) ,
xlab="x value" , ylab="probability" )
for ( i in 1:10 ) lines( xseq , p[i,] , lwd=3 ,
col=2 )
```

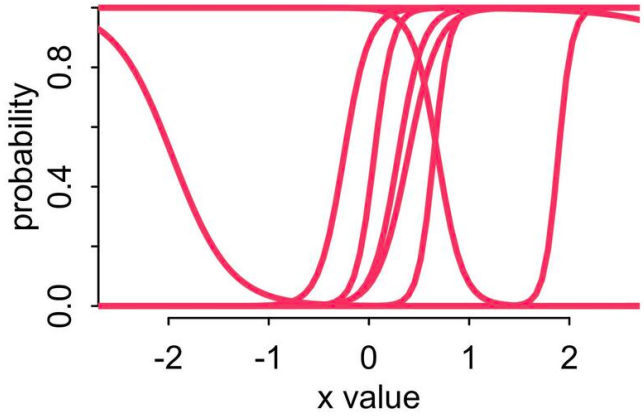


Logistic priors

$\alpha \sim \text{Normal}(0,1.5)$
$\beta \sim \text{Normal}(0,0.5)$



$\alpha \sim \text{Normal}(0,10)$
$\beta \sim \text{Normal}(0,10)$



Aggregated binomial example: UC Berkeley Admissions

- 4526 graduate school applications for 1973 UC Berkeley
- Stratified by department and gender of applicant
- See `?UCBAdmissions` for details and citations

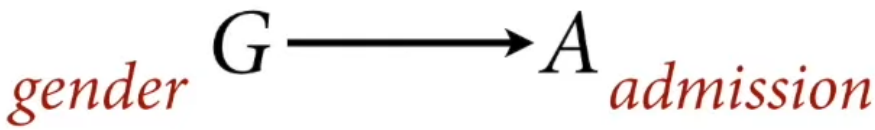
- Evidence of gender discrimination?

	dept	admit	reject	applications	gender
1	A	512	313	825	male
2	A	89	19	108	female
3	B	353	207	560	male
4	B	17	8	25	female
5	C	120	205	325	male
6	C	202	391	593	female
7	D	138	279	417	male
8	D	131	244	375	female
9	E	53	138	191	male
10	E	94	299	393	female
11	F	22	351	373	male
12	F	24	317	341	female

Admissions: drawing the owl

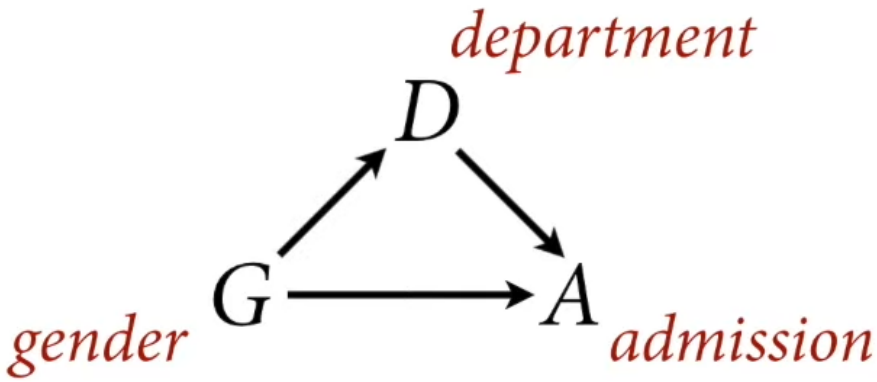
1. Estimand(s) - already said in a vague way
2. Scientific model(s) - will help precisely defining the estimand
3. Statistical model(s)
4. Validate and analyse

Admissions: Estimand and scientific models together



Discrimination is a hard question. Instead, we will ask of gender influences admission.

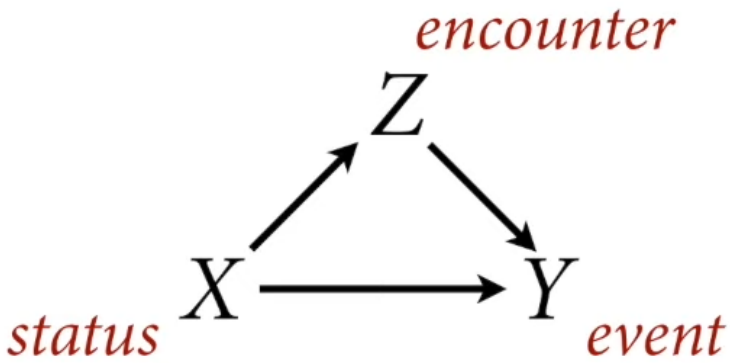
Admissions: Estimand and scientific models together



There is another pathway; some departments are very selective, others would accept almost everyone.

Total vs direct causal effect

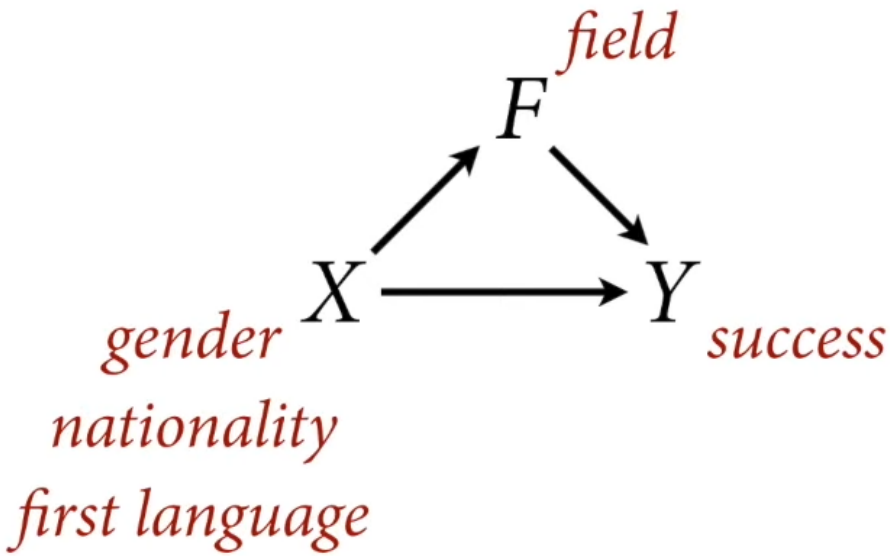
Encounters & Discrimination



A very common causal structure. Status is Exposure (target of discrimination). Direct and indirect influence on the outcome event.

Encounters & Discrimination

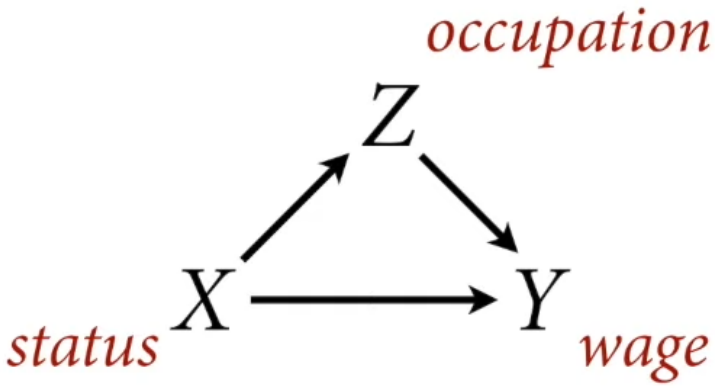
Grant application example



The UC Berkeley Admission problem is interesting per se, but we learn a lot about the many similar situations.

Encounters & Discrimination

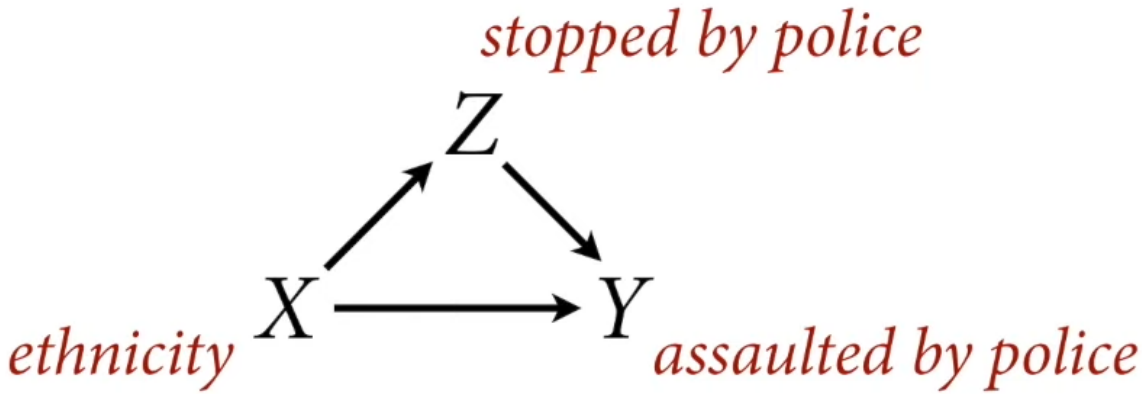
Wage example



The UC Berkeley Admission problem is interesting per se, but we learn a lot about the many similar situations.

Encounters & Discrimination

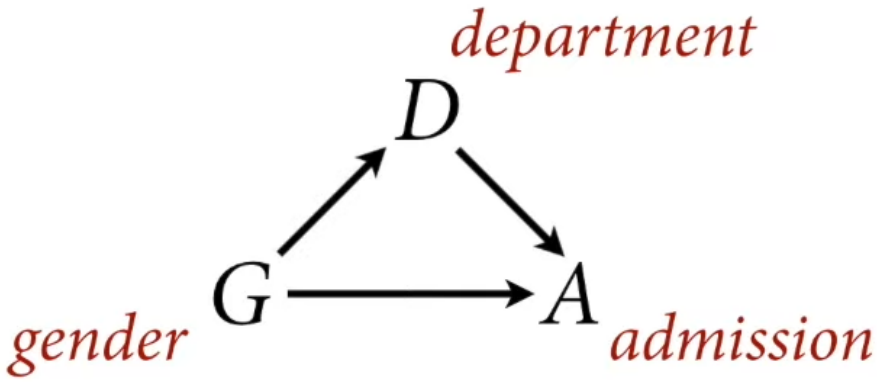
Policing example



The UC Berkeley Admission problem is interesting per se, but we learn a lot about the many similar situations.

Admissions

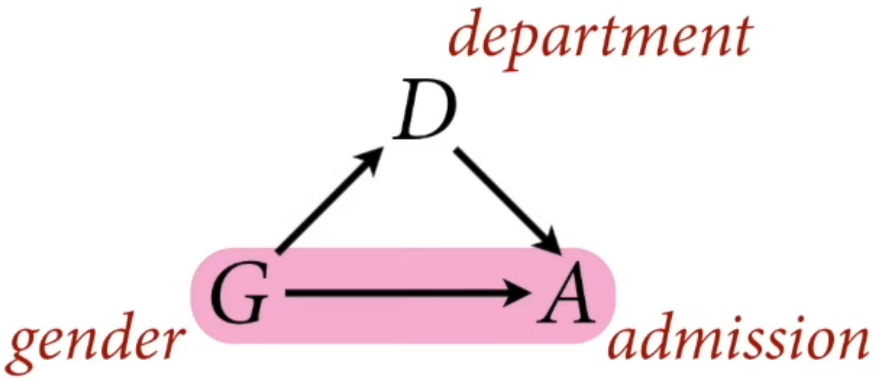
Which path is "discrimination"



- Argue that all are depending on the kind of discrimination you consider
- Again, the causes are not in the data
- The conceptual work will lead our thinking and conclusions.

Admissions

Which path is “discrimination”

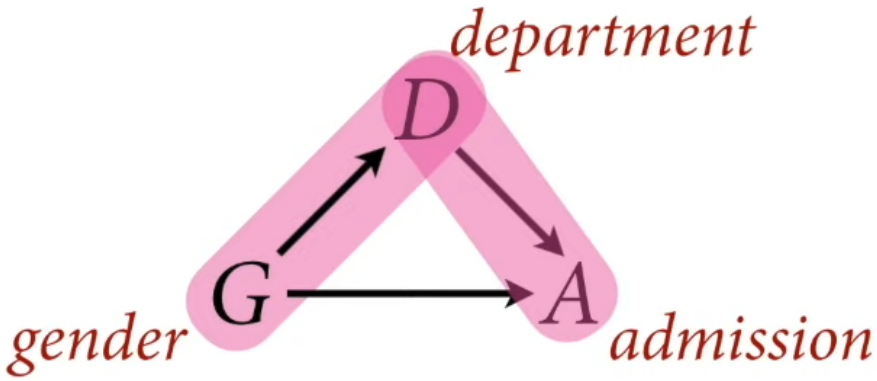


Direct discrimination

(status-based or taste-based discrimination, latter is difference, has got an inference mediator)

Admissions

Which path is "discrimination"



Indirect discrimination

(Structural discrimination, e.g. some gender don't apply to some departments; could lead to unequal access to education)

Generative model

- How can choice of department create structural discrimination?
- When department vary in baseline admission rates.

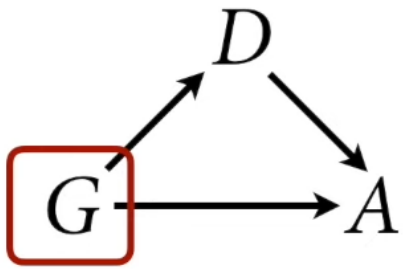
```
# basic mediator scenario
N <- 1000 # number of applicants
# even gender distribution
G <- sample(1:2, size=N, replace=TRUE)
# gender 1 tends to apply to dept 1,
# 2 to 2
D <- rbern(N, ifelse(G==1, 0.3, 0.8))+1
# [dept, gender]
accept_rate <- matrix(c(0.1,0.3,0.1,0.3)
# simulate acceptance
A <- rbern( N , accept_rate[D,G] )
accept_rate
```

```
##      [,1] [,2]
## [1,]  0.1  0.1
## [2,]  0.3  0.3
```

Generative model: gender simulation

```r
# generative model, basic mediator scenario

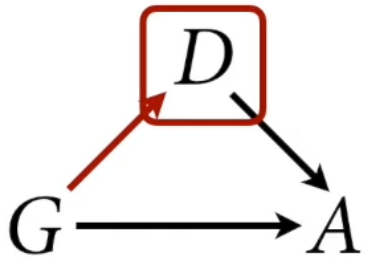
N <- 1000 # number of applicants
# even gender distribution
G <- sample( 1:2 , size=N , replace=TRUE )
# gender 1 tends to apply to department 1, 2 to 2
D <- rbern( N , ifelse( G==1 , 0.3 , 0.8 ) ) + 1
# matrix of acceptance rates [dept,gender]
accept_rate <- matrix( c(0.1,0.3,0.1,0.3) , nrow=2 )
# simulate acceptance
A <- rbern( N , accept_rate[D,G] )
```



Generative model: department distribution with gender preference

```r
# generative model, basic mediator scenario

N <- 1000 # number of applicants
# even gender distribution
G <- sample( 1:2 , size=N , replace=TRUE )
# gender 1 tends to apply to department 1, 2 to 2
D <- rbern( N , ifelse( G==1 , 0.3 , 0.8 ) ) + 1
# matrix of acceptance rates [dept,gender]
accept_rate <- matrix( c(0.1,0.3,0.1,0.3) , nrow=2 )
# simulate acceptance
A <- rbern( N , accept_rate[D,G] )
```

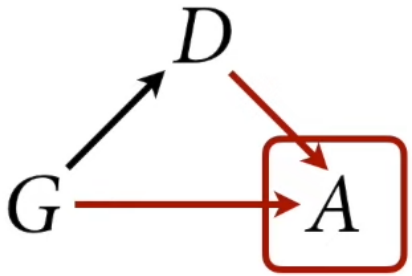


Generative model: Acceptance depends on G and A

```
# generative model, basic mediator scenario

N <- 1000 # number of applicants
# even gender distribution
G <- sample( 1:2 , size=N , replace=TRUE )
# gender 1 tends to apply to department 1, 2 to 2
D <- rbern( N , ifelse( G==1 , 0.3 , 0.8 ) ) + 1
# matrix of acceptance rates [dept,gender]
accept_rate <- matrix( c(0.1,0.3,0.1,0.3) , nrow=2 )
# simulate acceptance
A <- rbern( N , accept_rate[D,G] )
```

```
> accept_rate
     [,1] [,2]
[1,]  0.1  0.1
[2,]  0.3  0.3
```



(no discrimination here in `accept_rate`)

Admissions

What happens on acceptance rates?

```
# generative model, basic mediator scenario

N <- 1000 # number of applicants
# even gender distribution
G <- sample( 1:2 , size=N , replace=TRUE )
# gender 1 tends to apply to department 1, 2 to 2
D <- rbern( N , ifelse( G==1 , 0.3 , 0.8 ) ) + 1
# matrix of acceptance rates [dept,gender]
accept_rate <- matrix( c(0.1,0.3,0.1,0.3) , nrow=2 )
# simulate acceptance
A <- rbern( N , accept_rate[D,G] )
```

```
> table(G,D)
   D
G     1   2
  1 361 161
  2  99 379
```

```
> table(G,A)
   A
G     0   1
  1 421 101
  2 350 128
```

Accept rates
Gender 1: 19%
Gender 2: 27%

Gender 1 is accepted less in the overall sample in this particular example.

Does that puzzle you?

Admissions

What happens on acceptance rates?

```
# generative model, basic mediator scenario

N <- 1000 # number of applicants
# even gender distribution
G <- sample( 1:2 , size=N , replace=TRUE )
# gender 1 tends to apply to department 1, 2 to 2
D <- rbern( N , ifelse( G==1 , 0.3 , 0.8 ) ) + 1
# matrix of acceptance rates [dept,gender]
accept_rate <- matrix( c(0.1,0.3,0.1,0.3) , nrow=2 )
# simulate acceptance
A <- rbern( N , accept_rate[D,G] )
```

```
> table(G,D)
   D
G     1   2
  1 361 161
  2  99 379
```

```
> table(G,A)
   A
G     0   1
  1 421 101
  2 350 128
```

Accept rates
Gender 1: 19%
Gender 2: 27%

Gender 1 is accepted less in the overall sample in this particular example.

Does that puzzle you?

The total causal effect via the structural avenue leads to gender acceptance imbalance.

Admissions: change the simulations

```
# generative model, basic mediator scenario

N <- 1000 # number of applicants
# even gender distribution
G <- sample( 1:2 , size=N , replace=TRUE )
# gender 1 tends to apply to department 1, 2 to 2
D <- rbern( N , ifelse( G==1 , 0.3 , 0.8 ) ) + 1
# matrix of acceptance rates [dept,gender]
accept_rate <- matrix( c(0.05,0.2,0.1,0.3) , nrow=2 )
# simulate acceptance
A <- rbern( N , accept_rate[D,G] )
```

```
> table(G,D)
   D
G     1   2
  1 355 164
  2  95 386
```

```
> table(G,A)
   A
G     0   1
  1 473  46
  2 404  77
```

```
> accept_rate
     [,1] [,2]
[1,] 0.05  0.1
[2,] 0.20  0.3
```

Accept rates
Gender 1: 9%
Gender 2: 16%

Gender 1 is still accepted less in the overall sample in this 2nd example.

Same pattern, just different (smaller) numbers.

Analyse what is happening

- The resulting acceptance rates are a start, but lots of the possible stories are missing
- Admission rates usually depend on application pool size and distribution of qualifications
- In principle, we should sample applicant pool and then sort (ranking can contain 'errors') to select admissions
- Rates are conditional on structure of applicant pool

```
> table(G,D)
   D
G     1   2
  1 355 164
  2  95 386
```

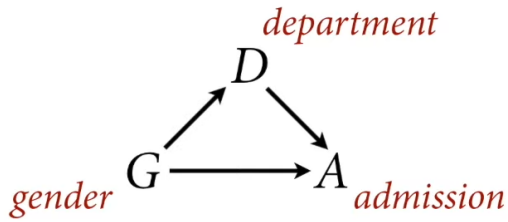
```
> table(G,A)
   A
G     0   1
  1 473  46
  2 404  77
```

Accept rates
Gender 1: 9%
Gender 2: 16%

A statistical model

```
# generative model, basic mediator scenario

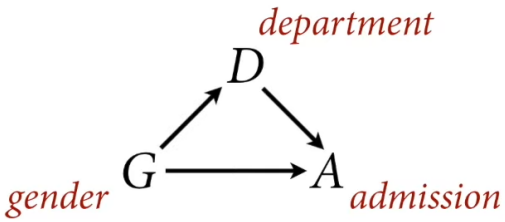
N <- 1000 # number of applicants
# even gender distribution
G <- sample( 1:2 , size=N , replace=TRUE )
# gender 1 tends to apply to department 1, 2 to 2
D <- rbern( N , ifelse( G==1 , 0.3 , 0.8 ) ) + 1
# matrix of acceptance rates [dept,gender]
accept_rate <- matrix( c(0.05,0.2,0.1,0.3) , nrow=2 )
# simulate acceptance
A <- rbern( N , accept_rate[D,G] )
```



Estimand: total effect of G

$$A_i \sim \text{Bernoulli}(p_i)$$
$$\text{logit}(p_i) = \alpha[G_i]$$
$$\alpha = \overbrace{[\alpha_1, \alpha_2]}^{\textit{Genders}}$$

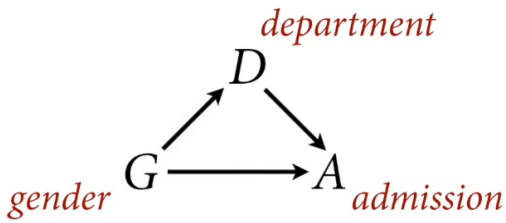
$$P(A_i = 1) = p_i$$
$$p_i = \frac{\exp(\alpha[G_i])}{1+\exp(\alpha[G_i])}$$



Estimand: total effect of G

$$A_i \sim \text{Bernoulli}(p_i)$$
$$\text{logit}(p_i) = \alpha[G_i, D_i]$$

$$\alpha = \overbrace{\begin{pmatrix} \alpha_{1,1} & \alpha_{1,2} \\ \alpha_{2,1} & \alpha_{2,2} \end{pmatrix}}^{\textit{Departments}} \textit{Genders}$$



Discrimitation?

Two models

Total Effect

$A_i \sim \text{Bernoulli}(p_i)$ $\text{logit}(p_i) = \alpha[G_i]$

$\alpha_j \sim \text{Normal}(0, 1)$

Direct Effect(s)

$A_i \sim \text{Bernoulli}(p_i)$

$\text{logit}(p_i) = \alpha[G_i, D_i]$

$\alpha_{j,k} \sim \text{Normal}(0, 1)$

Discrimitation?

Two models

Total Effect

$$A_i \sim \text{Bernoulli}(p_i) \; \text{logit}(p_i) = \alpha[G_i]$$
$$\alpha_j \sim \text{Normal}(0, 1)$$

```
dat_sim <- list( A=A , D=D , G=G )

m1 <- ulam(
    alist(
        A ~ bernoulli(p),
        logit(p) <- a[G],
        a[G] ~ normal(0,1)
    ), data=dat_sim , chains=4 , cores=4 )
```

Direct Effect(s)

$$A_i \sim \text{Bernoulli}(p_i)$$
$$\text{logit}(p_i) = \alpha[G_i, D_i]$$
$$\alpha_{j,k} \sim \text{Normal}(0, 1)$$

```
dat_sim <- list( A=A , D=D , G=G )

m2 <- ulam(
    alist(
        A ~ bernoulli(p),
        logit(p) <- a[G,D],
        matrix[G,D]:a ~ normal(0,1)
    ), data=dat_sim , chains=4 , cores=4 )
```

Summary

What we know so far for GLM:

- Outcome distributions
- Link function
- Logistic regression

Tomorrow:

GLMs in practice